



School of Computer Science

COMP47470: Big Data Programming

Lab practical MySQL2 Solutions

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MySQL Component 1

Problem 1: Solution:

```
CREATE TABLE GlobalWeatherRepository (  
    country VARCHAR(64) NOT NULL,  
    location_name VARCHAR(64) NOT NULL,  
    latitude DOUBLE NOT NULL,  
    longitude DOUBLE NOT NULL,  
    timezone VARCHAR(64) NOT NULL,  
    last_updated_epoch BIGINT NOT NULL,  
    last_updated DATETIME(0) NOT NULL,  
    temperature_celsius DOUBLE NOT NULL,  
    temperature_fahrenheit DOUBLE NOT NULL,  
    condition_text VARCHAR(64) NOT NULL,  
    wind_mph DOUBLE NOT NULL,  
    wind_kph DOUBLE NOT NULL,  
    wind_degree SMALLINT NOT NULL,  
    wind_direction CHAR(4) NOT NULL,  
    pressure_mb FLOAT NOT NULL,  
    pressure_in FLOAT NOT NULL,  
    precip_mm FLOAT NOT NULL,  
    precip_in FLOAT NOT NULL,  
    humidity SMALLINT NOT NULL,  
    cloud SMALLINT NOT NULL,  
    feels_like_celsius FLOAT NOT NULL,  
    feels_like_fahrenheit FLOAT NOT NULL,  
    visibility_km FLOAT NOT NULL,  
    visibility_miles FLOAT NOT NULL,  
    uv_index FLOAT NOT NULL,  
    gust_mph FLOAT NOT NULL,  
    gust_kph FLOAT NOT NULL,  
    air_quality_Carbon_Monoxide FLOAT NOT NULL,  
    air_quality_Ozone FLOAT NOT NULL,  
    air_quality_Nitrogen_dioxide FLOAT NOT NULL,  
    air_quality_Sulphur_dioxide FLOAT NOT NULL,  
    air_quality_PM2point5 FLOAT NOT NULL,  
    air_quality_PM10 FLOAT NOT NULL,  
    air_quality_us_epa_index SMALLINT NOT NULL,  
    air_quality_gb_defra_index SMALLINT NOT NULL,  
    sunrise VARCHAR(64) NOT NULL,  
    sunset VARCHAR(64) NOT NULL,
```

```
        moonrise VARCHAR(64) NOT NULL,  
        moonset VARCHAR(64) NOT NULL,  
        moon_phase VARCHAR(64) NOT NULL,  
        moon_illumination SMALLINT NOT NULL,  
PRIMARY KEY (country,location_name,latitude,longitude,last_updated_epoch)  
);
```

Problem 2: Solution:

```
LOAD DATA LOCAL INFILE 'GlobalWeatherRepository.csv' INTO TABLE  
GlobalWeatherRepository FIELDS TERMINATED BY ',' IGNORE 1 LINES;
```

Output:

Query OK, 26782 rows affected (0.38 sec)
Records: 26782 Deleted: 0 Skipped: 0 Warnings: 0

Problem 3: Solution:

```
SELECT      country,  
            location_name,  
            last_updated,  
            temperature_celsius  
FROM GlobalWeatherRepository  
WHERE temperature_celsius =  
        (SELECT MAX(temperature_celsius)  
         FROM GlobalWeatherRepository);
```

Output:

country	location_name	last_updated	temperature_celsius
Kuwait	Kuwait City	2024-06-19 16:45:00	49.2

Problem 4: Solution:

Following are valid solutions:

```
SELECT      country,  
            location_name,
```

```

        Last_updated,
        air_quality_PM10
FROM GlobalWeatherRepository
WHERE country = 'Ireland'
      AND location_name = 'Dublin'
      AND air_quality_PM10 =
          (SELECT MAX(air_quality_PM10)
           FROM GlobalWeatherRepository
           WHERE country = 'Ireland' AND location_name = 'Dublin');
OR

```

```

SELECT    country,
          location_name,
          last_updated,
          air_quality_PM10
FROM GlobalWeatherRepository
WHERE location_name = 'Dublin'
ORDER BY air_quality_PM10 DESC LIMIT 1;

```

Output:

country	location_name	last_updated	air_quality_PM10
Ireland	Dublin	2024-09-07 13:00:00	61.605

Problem 5: Solution:

```

SELECT    country,
          location_name,
          Last_updated,
          air_quality_PM2point5
FROM GlobalWeatherRepository
ORDER BY air_quality_PM2point5 ASC LIMIT
1;

```

Output:

country	location_name	last_updated	air_quality_PM2point5
Andorra	Andorra La Vella	2024-05-17 18:00:00	0.5

MySQL Component 2

Problem 1: Solution:

```
CREATE TABLE STATIONS (
    StationID INT NOT NULL,
    Name VARCHAR(64) NOT NULL,
    Latitude DOUBLE NOT NULL,
    Longitude DOUBLE NOT NULL,
    Dockcount INT NOT NULL,
    Landmark VARCHAR(64) NOT NULL,
    Installation DATE NOT NULL,
    PRIMARY KEY (StationID) );
```

```
CREATE TABLE TRIPS (
    TripID INT NOT NULL,
    Duration INT NOT NULL,
    StartDate DATE NOT NULL,
    StartStation VARCHAR(64) NOT NULL,
    StartTerminal INT NOT NULL,
    EndDate DATE NOT NULL,
    EndStation VARCHAR(64) NOT NULL,
    EndTerminal INT NOT NULL,
    BikeNo INT NOT NULL,
    SubscriberType VARCHAR(64) NOT NULL,
    ZipCode INT NOT NULL,
    PRIMARY KEY (TripID),
    FOREIGN KEY (StartTerminal) REFERENCES STATIONS (StationID),
    FOREIGN KEY (EndTerminal) REFERENCES STATIONS (StationID) );
```

Problem 2: Solution:

```
LOAD DATA LOCAL INFILE '201508_station_data.csv' INTO TABLE STATIONS
FIELDS TERMINATED BY ',' IGNORE 1 LINES
(StationID,Name,Latitude,Longitude,Dockcount,Landmark,@Installation)
```

```
set Installation = str_to_date(@Installation, '%m/%d/%Y');
```

```
LOAD DATA LOCAL INFILE '201508_trip_data_zipcode_0.csv' INTO TABLE TRIPS  
FIELDS TERMINATED BY ';' IGNORE 1 LINES
```

```
(TripID,Duration,@StartDate,StartStation,StartTerminal,@EndDate,EndStation,End  
Terminal,BikeNo,SubscriberType,ZipCode)
```

```
set StartDate = str_to_date(@StartDate, '%m/%d/%Y'),
```

```
    EndDate = str_to_date(@EndDate, '%m/%d/%Y');
```

Problem 3: Solution:

```
SELECT      StartStation,  
            EndStation,  
            Duration
```

```
FROM TRIPS
```

```
WHERE Duration < (SELECT MAX(Duration) FROM TRIPS)
```

```
ORDER BY Duration DESC LIMIT 1
```

Output:

StartStation	EndStation	Duration
Market at Sansome	Yerba Buena Center of the Arts (3rd @ Howard)	2137000

Problem 4: Solution:

```
SELECT      t1.Name,  
            t1.Dockcount,  
            t2.EndStation,  
            t2.Duration
```

```
FROM STATIONS as t1 LEFT JOIN TRIPS as t2
```

```
    ON t1.StationID = t2.StartTerminal
```

```
WHERE t2.Duration > 1000000;
```

Output:

Name	EndStation	Dockcount	Duration
South Van Ness at Market	2nd at Folsom	19	17270400
San Antonio Shopping Center	Castro Street and El Camino Real	15	1852590
Market at Sansome	Yerba Buena Center of the Arts (3rd @ Howard)	27	2137000
University and Emerson	University and Emerson	11	1133540

Problem 5: Solution:

```

SELECT      t1.Name,
            t1.Dockcount,
            t2.EndStation,
            t2.Duration
FROM STATIONS as t1 RIGHT JOIN TRIPS as t2
  ON t1.StationID = t2.StartTerminal
WHERE t2.Duration > 1000000;

```

Output:

Name	EndStation	Dockcount	Duration
South Van Ness at Market	2nd at Folsom	19	17270400
San Antonio Shopping Center	Castro Street and El Camino Real	15	1852590
Market at Sansome	Yerba Buena Center of the Arts (3rd @ Howard)	27	2137000
University and Emerson	University and Emerson	11	1133540

Note the order of the select attributes (Name, Dockcount, EndStation, and Duration) do not matter. Furthermore, the prefixes (t1 and t2) are not necessary for disambiguation.